

Finite Abelian Power Maps: Complete Functional Graphs, Zeta Products, and Koopman Jordan Atlases

Route-A source-local certificate HCS-C264

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Abstract

For every finite abelian group G and integer $d \geq 1$, we classify the entire functional graph of $g \mapsto g^d$. A canonical prime-support split separates a periodic automorphism from a nilpotent transient factor. It gives all fixed and primitive counts and the complete finite dynamical zeta product. Every periodic vertex receives the same explicitly specified rooted in-tree, including every tail layer and its saturation height; the singular constant map $d = 0$ is closed separately. For composition on the full function space we further determine every power rank, every zero Jordan-block size, and the characteristic polynomial. The theorem is parameter-uniform; exact finite enumeration is used only as a replayable regression certificate.

1 Prime-support splitting

Write the group additively as

$$G = \bigoplus_{i=1}^r C_{n_i}, \quad T_d(x) = dx.$$

For $d \geq 1$, factor $n_i = a_i b_i$ so that $(a_i, d) = 1$ and every prime divisor of b_i divides d . Thus $(a_i, b_i) = 1$. Put

$$A = \bigoplus_i C_{a_i}, \quad B = \bigoplus_i C_{b_i}.$$

The Chinese remainder theorem identifies $G = A \times B$. Multiplication by d is an automorphism on A , whereas every element of B reaches zero. Functional graphs of power maps over finite groups provide the established lineage for this decomposition [1]; the theorem below is proved locally, and no literature-priority claim is made.

Theorem 1 (complete cycle atlas). *The periodic set of T_d is exactly $A \times \{0\}$. For every $n \geq 1$,*

$$F_n := |\text{Fix}(T_d^n)| = \prod_i \gcd(d^n - 1, a_i). \quad (1)$$

The numbers of exact-period points and cycles are respectively

$$P_m = \sum_{e|m} \mu(m/e) F_e, \quad C_m = P_m/m. \quad (2)$$

In particular, only finitely many C_m are nonzero and

$$\zeta_{T_d}(t) := \exp\left(\sum_{n \geq 1} F_n \frac{t^n}{n}\right) = \prod_{m \geq 1} (1 - t^m)^{-C_m}. \quad (3)$$

Proof. If (a, b) is periodic, then $d^k b = b$ for some $k > 0$. But a power of d annihilates B , so $b = 0$; conversely the finite automorphism on A makes every $(a, 0)$ periodic. On C_{a_i} the equation $(d^n - 1)x = 0$ has $\gcd(d^n - 1, a_i)$ solutions, proving (1). Partitioning F_m by exact period and applying Möbius inversion gives (2). Expanding $-\log(1 - t^m)$ and collecting the coefficient of t^n/n proves (3). $\square$

2 Uniform transient trees and boundary faces

Define

$$K_j = \left| \ker(T_d^j|_B) \right| = \prod_i \gcd(d^j, b_i), \quad K_0 = 1.$$

If $b \in B$ first reaches zero after j steps, then its exact tail is j . For each periodic vertex, invertibility on A supplies a unique A -coordinate at every backward step. Hence its attached rooted tree is canonically the tree of $b \mapsto db$ on B , and its depth- j population is

$$L_j = K_j - K_{j-1} \quad (j \geq 1), \quad L_0 = 1. \quad (4)$$

This specifies the complete graph: cycles are given by (2), and every cycle vertex is decorated by the same explicitly defined B -tree. The maximal tail height is

$$h = \max_{p|d} \left\lceil \frac{\max_i v_p(b_i)}{v_p(d)} \right\rceil, \quad (5)$$

with an empty maximum interpreted as zero.

The identity face $d = 1$ lies inside the theorem: $A = G$, $B = 0$, every point is fixed, and there is no transient sector. The exponent $d = 0$ is kept outside the prime-support notation. It is the constant map to the identity, with one fixed point and $|G| - 1$ vertices of exact tail one; its source zeta is $(1 - t)^{-1}$.

3 Full-function Koopman Jordan atlas

Let $U : \mathbb{C}^G \rightarrow \mathbb{C}^G$ be the composition operator $(Uf)(x) = f(T_d x)$. Distinct rows of its zero-one matrix are indexed by the distinct values of T_d , so

$$\text{rank}(U^j) = R_j := |\text{im } T_d^j| = |A| \prod_i \frac{b_i}{\gcd(d^j, b_i)}. \quad (6)$$

The ranks stabilize at $|A|$. For a nilpotent Jordan block of size s , the rank loss from power $j - 1$ to j is one exactly when $s \geq j$. Taking a second difference therefore gives the number of zero blocks of exact size j :

$$Z_j = R_{j-1} - 2R_j + R_{j+1}. \quad (7)$$

The recurrent quotient is a disjoint union of the cycles counted in (2), so its permutation characteristic factors are $\lambda^m - 1$. Consequently the entire operator, including its transient algebraic multiplicity, satisfies

$$\det(\lambda I - U) = \lambda^{|G|-|A|} \prod_{m \geq 1} (\lambda^m - 1)^{C_m}. \quad (8)$$

For $d = 0$, $U^2 = U$; it is diagonalizable with eigenvalues 1 once and zero $|G| - 1$ times. In general U need not be normal or unitary; only the bijective face $B = 0$ is a permutation operator.

4 Exact certificate and Route-A boundary

The frozen corpus crosses 34 invariant-factor group types with $d = 0, \dots, 18$: 646 maps and 21,280 directly enumerated case-elements. It stores 1,320 fixed-point cells, 933 cycle factors, 1,132 tail layers, and 485 nonzero zero-Jordan entries. An independent implementation closes 202,656 assertions without importing the producer. SymPy closes 1,029 exact characteristic-polynomial and rank checks on 220 full matrices; fresh byte replay passes, and repaired-hash mutations are rejected 33/33. These are regression receipts, not a finite-census proof of the theorem.

This certificate concerns a finite source dynamical system. It supplies no arithmetic local datum, bad Euler factor, root number, automorphy statement, target divisor, functional equation, or Hilbert–Pólya operator. Under `NO_BAD_EULER_OR_ROOT_NUMBER`, its frozen Route-A tuple is

$$(\text{AO_WEAK_ARITHMETIC_RELATION}, \text{A1_PASS_ANALYTIC}, \text{A2_FAIL}, \\ \text{A3_FAIL}, \text{A4_FORMAL_HINT}).$$

The verdict is `ROUTE_A_PARTIAL`; Route B is disabled. The Koopman matrix is a formal source operator, not a target spectral realization. We claim ownership only of this workspace certificate, not literature priority.

References

- [1] C. Qureshi and L. Reis, *On the functional graph of the power map over finite groups*, *Discrete Mathematics* **346** (2023), 113393, [doi:10.1016/j.disc.2023.113393](https://doi.org/10.1016/j.disc.2023.113393).